To: Dr. Kristina Hostáková, kristina.hostakova@inf.ethz.ch
Subject: A fully direct proof using a reduction

Dear Dr. Hostáková,

Below is a fully direct proof that IND-CPA security implies semantic security, using a reduction. I also include the alternative final step by contraposition from Lecture 2, alongside the direct conclusion, to compare the two presentations.

I previously had the impression that using a reduction necessarily introduced a contrapositive step in a proof. Writing out the argument below helped me understand the distinction. My understanding is that a reduction constructs an algorithm using another algorithm as a subroutine, providing the interaction that the subroutine expects. Whether we use the resulting implication directly or by contraposition is a separate choice in presenting the proof.

Could you let me know whether this understanding is correct?

Best regards,
Klevis

Theorem 1.14 (Semantic security $\iff$ IND-CPA)**.**

$$\text{SemanticSecure}(\Pi) \iff \text{IND-CPA-Secure}(\Pi).$$

$$\Pi : \textit{ public key encryption scheme.}$$

By direct proof: IND-CPA $\implies$ semantic security.

$$\text{Goal:} \quad \begin{bmatrix} \forall \text{ PPT } D\, \forall \lambda \in \mathbb{N} : \\ \text{Adv}^{\text{IND-CPA}}_{\Pi,D}(\lambda) \le \text{negl}(\lambda) \end{bmatrix} \implies \begin{bmatrix} \forall \text{ PPT } A\, \exists \text{ PPT } A'\, \forall \text{ PPT } \mathsf{Samp}\, \forall \text{ PT } f : \\ \forall \lambda \in \mathbb{N} : \quad |\epsilon_A(\lambda) - \epsilon_{A'}(\lambda)| \le \text{negl}(\lambda) \end{bmatrix}.$$

$$\text{Idea:} \quad \begin{bmatrix} \exists \text{ PPT } D\, \exists \text{ polynomial } p\, \forall \lambda \in \mathbb{N} : \\ |\epsilon_A(\lambda) - \epsilon_{A'}(\lambda)| \le p(\lambda)\, \text{Adv}^{\text{IND-CPA}}_{\Pi,D}(\lambda) + \text{negl}(\lambda) \end{bmatrix} \overset{\text{IND-CPA-Secure}(\Pi)}{\implies} |\epsilon_A - \epsilon_{A'}| \text{ negligible.}$$

$$\forall \text{ PPT } D : \quad \underbrace{2\left| P^{\text{IND-CPA}}_{\Pi,D,\text{win}}(\lambda) - \frac{1}{2} \right|}_{\text{Adv}^{\text{IND-CPA}}_{\Pi,D}(\lambda)} \overset{\text{IND-CPA-Secure}(\Pi)}{\le} \text{negl}(\lambda). \tag{2}$$

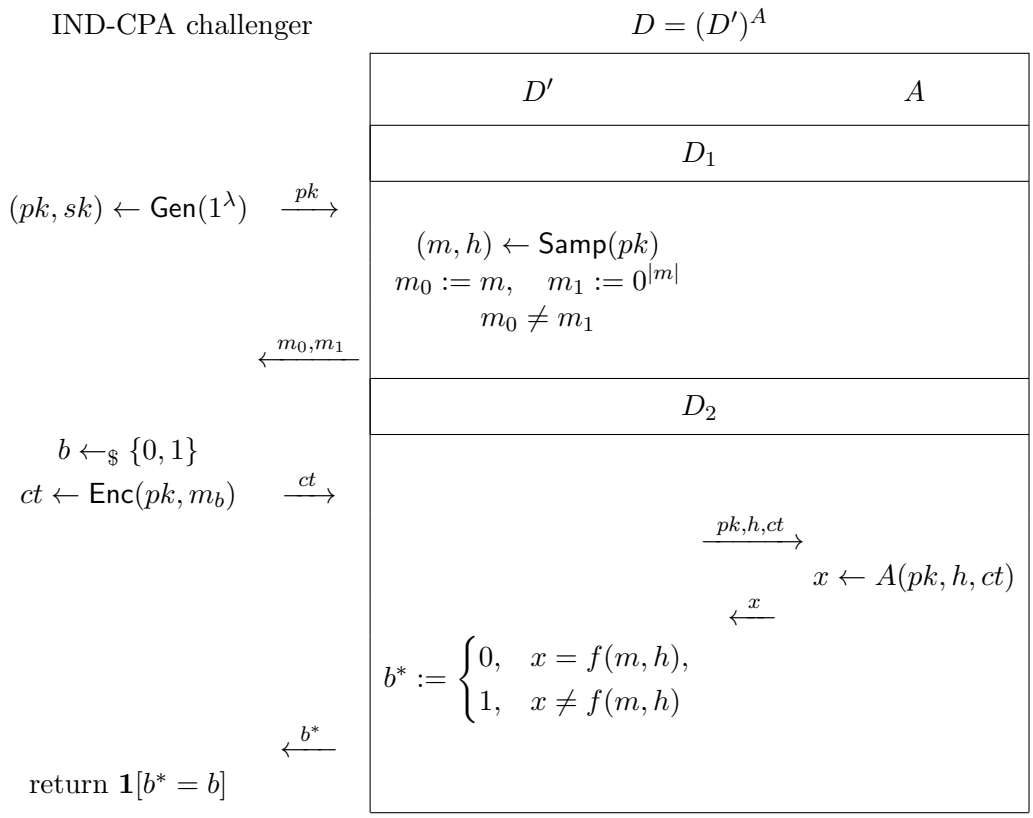
Idea 2: Construct the IND-CPA game for Idea 1.

We want to construct an IND-CPA attacker D so that both ϵ_A and $\epsilon_{A'}$ appear in its winning probability. This would let us relate their difference to D's advantage, as required by Idea 1. The IND-CPA game already provides a challenge bit b with two possible values. We use these two branches to include both executions in the same game. When $b = 0$, this runs A on an encryption of m, giving the real execution. When $b = 1$, it runs A on an encryption of $0^{|m|}$, which is exactly how A' is defined below. Thus the probabilities of correctly computing $f(m, h)$ in the two branches are ϵ_A and $\epsilon_{A'}$. We can then use them to calculate D's winning probability and advantage in this one IND-CPA game, and thereby bound the semantic-security gap.

$$\forall \text{ PPT } A : \quad A'(pk, h, \ell) := A(pk, h, \mathsf{Enc}(pk, 0^\ell)).$$

A' depends only on A and Π,

$\forall$ PPT Samp $\forall$ PT f : $\quad D = (D')^A$ uses the chosen Samp and f.



$$\begin{aligned}
\Pr[D \text{ wins} \mid b = 0] &= \Pr[A(pk, h, \mathsf{Enc}(pk, m)) = f(m, h)] \\
&= \epsilon_A(\lambda), \\
\Pr[D \text{ wins} \mid b = 1] &= \Pr[A'(pk, h, |m|) \neq f(m, h)] \\
&= 1 - \Pr[A'(pk, h, |m|) = f(m, h)] \\
&= 1 - \epsilon_{A'}(\lambda). \\
P^{\text{IND-CPA}}_{\Pi,D,\text{win}}(\lambda) &= \Pr[D \text{ wins} \mid b = 0] \Pr[b = 0] \\
&\quad + \Pr[D \text{ wins} \mid b = 1] \Pr[b = 1] \\
&= \frac{1}{2}\epsilon_A(\lambda) + \frac{1}{2}(1 - \epsilon_{A'}(\lambda)) \\
&= \frac{1}{2} + \frac{1}{2}\big(\epsilon_A(\lambda) - \epsilon_{A'}(\lambda)\big).
\end{aligned}$$

Direct conclusion

$$\begin{aligned}
|\epsilon_A(\lambda) - \epsilon_{A'}(\lambda)| &= 2\left|P^{\text{IND-CPA}}_{\Pi,D,\text{win}}(\lambda) - \frac{1}{2}\right| \\
&\overset{(2)}{\leq} \text{negl}(\lambda).
\end{aligned}$$

By contraposition

$$\begin{aligned}
&|\epsilon_A(\lambda) - \epsilon_{A'}(\lambda)| \text{ non-negligible} \\
\implies \quad &\text{Adv}^{\text{IND-CPA}}_{\Pi,D}(\lambda) = 2\left|P^{\text{IND-CPA}}_{\Pi,D,\text{win}}(\lambda) - \frac{1}{2}\right| \\
&\qquad = |\epsilon_A(\lambda) - \epsilon_{A'}(\lambda)| \quad \text{non-negligible} \\
&\qquad \overset{\text{IND-CPA-Secure}(\Pi)}{\implies} \perp. \\
\implies \quad &|\epsilon_A(\lambda) - \epsilon_{A'}(\lambda)| \leq \text{negl}(\lambda).
\end{aligned}$$